

AdaGuard LeakScore Framework

Mathematical Formulas Reference

Overall LeakScore

The final LeakScore is a weighted average of three component scores:

$$\text{LeakScore} = (\alpha \cdot \text{Entropy} + \beta \cdot \text{Label} + \gamma \cdot \text{Empirical}) / (\alpha + \beta + \gamma)$$

- Default weights: $\alpha = \beta = \gamma = 1.0$
- All components and the final score are normalized to $[0, 1]$
- A higher LeakScore indicates greater gradient leakage risk

1. Entropy LeakScore

Measures how concentrated the gradient distribution is. Concentrated gradients are easier to invert, indicating higher privacy risk.

Preprocessing Pipeline

- Flatten gradient tensor into a 1-D vector
- L2-normalize: $g_{\text{norm}} = g / \|g\|_2$
- Standardize: $g_{\text{std}} = (g_{\text{norm}} - \mu) / \sigma$
- Bin into a histogram with $B = 50$ bins to obtain probability distribution p

Sub-metrics

Metric	Entropy Formula	Leakage Score
Shannon	$H = -\sum p \cdot \log(p)$	$1 - H / \log(B)$
Rényi ($\alpha = 2$)	$H = -\log(\sum p^2)$	$1 - H / \log(B)$
Min-Entropy	$H = -\log(\max(p))$	$1 - H / \log(B)$

A uniform distribution yields entropy = $\log(B)$, giving score 0 (low risk). A highly peaked distribution yields low entropy, giving score near 1 (high risk).

2. Label LeakScore

Measures how much class-label information is embedded in the gradients. Three sub-metrics capture different aspects of label leakage.

2a. GLMIP (Gradient-Label Mutual Information Proxy)

Computes the ratio of between-class to total gradient variance, analogous to Fisher's linear discriminant:

$$S_B = \sum_c |C_c| \cdot ||\mu_c - \mu||^2 \quad (\text{between-class scatter})$$

$$S_W = \sum_c \sum_{g \in C_c} ||g - \mu_c||^2 \quad (\text{within-class scatter})$$

$$\text{Score} = S_B / (S_B + S_W)$$

- μ_c = mean gradient for class c ; μ = global mean gradient
- Score near 1 = gradients are highly class-separable (high label leakage)

2b. Confidence Gap

Measures prediction certainty from the model's softmax output:

$$\text{Score} = \max(\text{softmax}) - \text{second_max}(\text{softmax})$$

A large gap indicates high model confidence, which correlates with label-revealing gradient structure.

2c. Cosine Similarity

Measures how distinguishable gradients are across classes:

m = mean of all pairwise cosine similarities between class gradient means

$$\text{Score} = 1 - (m + 1) / 2$$

Low inter-class similarity maps to a high score, indicating an attacker can separate class signals from the gradient.

3. Empirical LeakScore

Directly measures leakage by attempting to reconstruct training data from gradients. Three gradient inversion attacks are run, each optimizing a dummy input to match the observed gradient.

Common Scoring Formula

$$\text{score} = \max(0, 1 - ||g_{\text{real}} - g_{\text{recon}}||_2 / ||g_{\text{real}}||_2)$$

A score near 1 means the attack nearly perfectly matched the real gradient, implying successful data reconstruction.

3a. GradInversion

$$\text{loss} = (1 - \text{cos_sim}(g_{\text{recon}}, g_{\text{real}})) + \lambda_{\text{tv}} \cdot \text{TV}(x)$$

TV(x) is total variation regularization, encouraging smooth reconstructions.

3b. GI-NAS

$$\text{loss} = ||g_{\text{recon}} - g_{\text{real}}||_2^2 + \lambda_{\text{gl}} \cdot \text{Var}(x.\text{mean}(\text{dim}=(2,3)))$$

Uses multi-restart optimization (default 3 restarts) and keeps the best result.

3c. GGCDM

$$\text{loss} = (1 - \text{cos_sim}(g_{\text{recon}}, g_{\text{real}})) + 0.01 \cdot ||g_{\text{recon}} - g_{\text{real}}||_2^2$$

Injects annealed noise: $x += (1 - \text{step}/n_{\text{iter}}) \cdot 0.01 \cdot \text{noise}$, which decays to zero over the optimization.

$$\text{Empirical LeakScore} = \text{mean}(\text{GradInversion}, \text{GI-NAS}, \text{GGCDM})$$

4. Fisher Information

Identifies which model weights carry the most information about training data. Used to decide which weights to encrypt via selective homomorphic encryption.

Per-weight Fisher (Empirical Approximation)

$$F_i = g_i^2$$

Average Fisher (Per Parameter)

$$f_{\text{round}} = \sum F_i / n \quad \text{where } n = \text{total number of parameters}$$

Normalized Fisher (Sigmoid Mapping)

$$f_{\text{norm}} = f_{\text{round}} / (f_{\text{round}} + 1)$$

Fisher Concentration (Gini Coefficient)

$$F_i = F_i / \sum F_i \quad (\text{normalize to distribution})$$

Sort values ascending: $s_1, s_2, \dots, s_n$

$$\text{Gini} = (2 \cdot \sum(i \cdot s_i)) / (n \cdot \sum s_i) - (n + 1) / n$$

Gini near 1 means Fisher information is concentrated in a small fraction of weights, making selective encryption efficient.

Encryption mask: The top `enc_pct%` of weights ranked by F_i are selected for homomorphic encryption.

5. MaskCrypt Vulnerability

From the MaskCrypt paper (Hu & Li, 2025). Measures per-weight vulnerability by combining gradient magnitude with the change in exposed model weights across rounds.

Per-weight Vulnerability

$$v[i] = g[i] \cdot (w_{\text{exposed}}[i] - w_{\text{trained}}[i])$$

- w_{exposed} = global weights exposed to the network from the previous round
- w_{trained} = locally trained weights in the current round

Vulnerability Score (Top-fraction Concentration)

$$\text{Score} = \Sigma(\text{top_k } |v|) / \Sigma(\text{all } |v|)$$

where `top_k` selects the `enc_pct%` of weights with the largest $|v|$. A high score means vulnerability is concentrated in a few weights.

L2 Score (Dispersion Measure)

$$\text{Score} = ||v||_2 / (\sqrt{n} \cdot \max |v|)$$

Encryption mask: Weights where $|v|$ exceeds the top `enc_pct%` threshold are selected for encryption.

6. Gradient Magnitude

A fast metric capturing the overall scale of gradients. Large gradients carry more information and are more susceptible to inversion.

Per-layer L2 Norm

$$\text{per_layer}[\text{name}] = ||g_{\text{name}}||_2$$

Total L2 Norm

$$L2 = \sqrt{(\Sigma \text{ per_layer_norm}^2)}$$

Magnitude Score (Smooth Normalization)

$$\text{Score} = \min(1, L2 / (L2 + 1))$$

Maps $[0, \infty)$ smoothly onto $[0, 1)$. A score near 1 indicates very large gradients with high information content.